

Physics-Informed ADMM for Path-Constrained Optimal Control via Indirect Method

Eunbyeol Cho¹

Dohoon Lee²

Jong-Han Kim^{2,†}

Abstract

This paper proposes a physics-informed alternating direction method of multipliers (PI-ADMM) framework for constrained continuous-time optimal control and applies it to a 3-DoF powered descent guidance (PDG) problem. The method combines an indirect primal update based on Pontryagin's Minimum Principle, derived from the augmented Hamiltonian system, with an outer-loop ADMM scheme for path-constraint enforcement. The primal trajectories are parameterized using Chebyshev-based constrained expressions and computed by minimizing the augmented PMP residuals. A fixed-point analysis establishes consistency with the first-order stationarity conditions of the original problem under the assumptions of augmented PMP satisfaction, convexity, and regularity. On the tested PDG benchmark, PI-ADMM achieves solution quality comparable to IPOPT and LCvx, with the lowest mean solve time across most tested node counts.

1 Introduction

Path-constrained continuous-time optimal control problems arise in many applications, including powered descent guidance (PDG), where one seeks dynamically feasible and constraint-satisfying trajectories under limited computation. In practical settings, the solver may need to operate in an online replanning loop while handling nonlinear dynamics and state and control constraints. This motivates computationally efficient methods that preserve continuous-time optimality structure while enforcing path feasibility effectively.

Existing approaches are broadly classified as direct and indirect. Direct methods transcribe the continuous-time problem into a nonlinear or convex program, making path constraints relatively straightforward to impose at discretization nodes [1–5], but often at the cost of large-scale

transcription, successive convexification, or computationally intensive solvers. Indirect methods instead apply Pontryagin's Minimum Principle (PMP) to derive a two-point boundary value problem (TPBVP) [6]; they preserve the continuous-time optimality structure and can yield highly accurate solutions, but are often sensitive to the initial costate guess and less convenient for complicated path constraints. These limitations have motivated recent interest in physics-informed indirect solvers.

Recently, physics-informed neural approaches [7] have been explored for solving TPBVPs arising from indirect optimal control. In this line of research, the Theory of Functional Connections (TFC) provides constrained expressions that analytically satisfy boundary conditions, whereas its neural extension, the Extreme Theory of Functional Connections (X-TFC), employs a single-layer neural network trained via the Extreme Learning Machine (ELM) algorithm to obtain computationally efficient continuous-time approximations [8–10]. These ideas have been applied to several optimal control problems [11–14]. However, earlier studies either focused on unconstrained problems, handled bounded controls through saturation and continuation, or did not provide a general treatment of state, control, and mixed state-control inequality path constraints. A recent extension, Physics-Informed Pontryagin Neural Networks (PoNNs) [15], addresses this gap by introducing additional Lagrange multipliers directly into the Hamiltonian and enforcing complementarity through the Fischer–Burmeister function. While this represents an important step toward constrained indirect neural optimal control, it requires optimizing the path multipliers and adding a complementarity residual term to the loss. Moreover, because the Fischer–Burmeister function is nonsmooth at the origin, its use may increase the tuning burden and complicate numerical treatment near active-set transitions.

This paper addresses these limitations by combining an indirect TPBVP solver with an outer-loop alternating direction method of multipliers (ADMM) framework. Specifically, the primal update retains the continuous-time PMP structure through an augmented Hamiltonian formulation, while path feasibility is enforced using projection-based ADMM updates. Based on this idea, we propose a *Physics-Informed ADMM* (PI-ADMM) framework for constrained continuous-time optimal control. The primal TPBVP is solved using a physics-informed architecture with Chebyshev-based constrained expressions [12, 16], which analytically satisfy boundary and transversality conditions, while the outer loop updates auxiliary and dual variables through operator splitting. In this way, the proposed method combines the continuous-time opti-

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¹Eunbyeol Cho is with the PGM Systems R&D Center at LG Defense&Aerospace Co., Ltd., Seongnam, South Korea. eunbyeol.cho@lgdna.com.

²Dohoon Lee and Jong-Han Kim are with the Department of Aerospace Engineering and the Program in Aerospace Systems Convergence, Inha University, Incheon, Republic of Korea. Jong-Han Kim is also with the Aerospace Systems Research Institute, Inha University. dohoonlee@inha.edu; jonghank@inha.ac.kr.

mality structure of indirect methods with the constraint-handling capability of first-order splitting methods.

The main contributions of this paper are summarized as follows:

- We propose a PI-ADMM framework that separates the augmented PMP-based primal solve from projection-based path-constraint handling.
- We develop a Chebyshev coefficient-space solver that analytically satisfies prescribed boundary conditions and retains a fixed LM system dimension under node refinement when the basis size is held fixed.
- A fixed-point analysis shows that exact fixed points of the continuous-time formulation satisfy the first-order stationarity conditions of the original problem provided that the convexity and regularity assumptions hold.

We evaluate PI-ADMM against IPOPT and LCvx in terms of solution quality and computational performance on a 3-DoF PDG benchmark.

2 Preliminaries

2.1 Continuous-Time Optimal Control Problem

We consider a continuous-time nonlinear optimal control problem with path constraints. Over $t \in [t_0, t_f]$, let $x(t) \in \mathbb{R}^{n_x}$ and $u(t) \in \mathbb{R}^{n_u}$ denote the state and control, respectively. The constrained OCP is given by (1), where Φ and $\mathcal{L}$ denote the terminal and running costs, f is the nonlinear dynamics, $g(t, x, u) \in \mathcal{C}$ denotes the path constraint, where $\mathcal{C}$ is a closed convex set.

$$\begin{aligned} \underset{x, u}{\text{minimize}} \quad & \Phi(t_f, x(t_f)) + \int_{t_0}^{t_f} \mathcal{L}(t, x(t), u(t)) dt \\ \text{subject to} \quad & \dot{x}(t) = f(t, x(t), u(t)), \\ & g(t, x(t), u(t)) \in \mathcal{C}, \\ & x(t_0) = x_0, \end{aligned} \quad (1)$$

To derive first-order necessary conditions, we introduce the Hamiltonian with costate $\lambda(t)$ and path multiplier $\mu(t)$:

$$H(t, x, u, \lambda, \mu) = \mathcal{L}(t, x, u) + \lambda^\top f(t, x, u) + \mu^\top g(t, x, u).$$

For normal extremals admitting an absolutely continuous costate and an L^2 path multiplier, the first-order stationarity conditions hold almost everywhere:

$$\begin{aligned} \dot{x} &= \frac{\partial H}{\partial \lambda} = f(t, x, u), \\ -\dot{\lambda} &= \frac{\partial H}{\partial x} = \frac{\partial \mathcal{L}}{\partial x} + \left(\frac{\partial f}{\partial x}\right)^\top \lambda + \left(\frac{\partial g}{\partial x}\right)^\top \mu, \\ 0 &= \frac{\partial H}{\partial u} = \frac{\partial \mathcal{L}}{\partial u} + \left(\frac{\partial f}{\partial u}\right)^\top \lambda + \left(\frac{\partial g}{\partial u}\right)^\top \mu. \end{aligned} \quad (2)$$

In addition, path feasibility and the associated multiplier condition are given by

$$g(t, x, u) \in \mathcal{C}, \quad \mu(t) \in \mathcal{N}_{\mathcal{C}}(g(t, x, u)), \quad (3)$$

where $\mathcal{N}_{\mathcal{C}}(\cdot)$ denotes the normal cone of $\mathcal{C}$. If $\mathcal{C} = \mathbb{R}^m$, this normal-cone condition is equivalent to the componentwise KKT conditions $g(t, x, u) \leq 0$, $\mu(t) \geq 0$, and $\mu(t) \odot g(t, x, u) = 0$. Together with the appropriate boundary and transversality conditions, these conditions form a TPBVP. However, coupled nonlinear path constraints make shooting-based indirect methods highly sensitive to the initial costate guess, motivating the constraint-handling strategy developed in the subsequent sections.

2.2 Alternating Direction Method of Multipliers

First-order methods handle constraints via gradient-based updates and proximal operations. Among them, ADMM is particularly effective when constraints can be isolated through auxiliary variables and projection steps.

Consider the constrained problem $\text{minimize}_{x \in \mathcal{C}} \phi(x)$. Introducing an auxiliary variable z yields the equivalent form (4), where $I_{\mathcal{C}}(z)$ is the indicator function of $\mathcal{C}$ (0 if $z \in \mathcal{C}$, $+\infty$ otherwise).

$$\begin{aligned} \underset{x, z}{\text{minimize}} \quad & \phi(x) + I_{\mathcal{C}}(z) \\ \text{subject to} \quad & x = z \end{aligned} \quad (4)$$

Using the scaled dual variable $y := \mu/\rho$, the augmented Lagrangian with penalty parameter $\rho > 0$ is given by

$$\mathcal{L}_\rho(x, z, y) = \phi(x) + I_{\mathcal{C}}(z) + \frac{\rho}{2} \|x - z + y\|^2 - \frac{\rho}{2} \|y\|^2 \quad (5)$$

ADMM then proceeds by alternating minimization and dual ascent, yielding the iterative updates:

$$\begin{aligned} x^{k+1} &= \arg \min_x \phi(x) + \frac{\rho}{2} \|x - z^k + y^k\|^2, \\ z^{k+1} &= \Pi_{\mathcal{C}}(x^{k+1} + y^k), \\ y^{k+1} &= y^k + x^{k+1} - z^{k+1}. \end{aligned} \quad (6)$$

In particular, the z -update reduces to the orthogonal projection $\Pi_{\mathcal{C}}(\cdot)$ onto the feasible set. This splitting-and-projection structure is used in this paper to handle coupled path constraints in nonlinear control problems.

3 Proposed Method: PI-ADMM

Our goal is to solve the continuous-time constrained OCP (1) using an indirect TPBVP approach while handling path constraints via an outer-loop splitting scheme.

3.1 Continuous-Time ADMM Framework

We formulate the ADMM splitting in the Hilbert space $L^2([t_0, t_f]; \mathbb{R}^m)$. Define the feasible trajectory set $\mathcal{Z} := \{z \in L^2([t_0, t_f]; \mathbb{R}^m) \mid z(t) \in \mathcal{C} \text{ a.e.}\}$. We introduce an auxiliary trajectory $z \in L^2([t_0, t_f]; \mathbb{R}^m)$ for the path constraint and enforce the coupling condition $g(t, x, u) - z(t) = 0$ almost everywhere on $[t_0, t_f]$. Path feasibility is imposed by requiring $z \in \mathcal{Z}$. This yields the following equivalent formulation:

$$\begin{aligned} \underset{x, u, z}{\text{minimize}} \quad & \Phi(t_f, x(t_f)) + \int_{t_0}^{t_f} [\mathcal{L}(t, x, u) + I_{\mathcal{C}}(z(t))] dt \\ \text{subject to} \quad & \dot{x} = f(t, x, u), \\ & g(t, x, u) - z(t) = 0, \\ & x(t_0) = x_0. \end{aligned}$$

where feasibility with respect to $\mathcal{C}$ is encoded through the indicator function $I_{\mathcal{C}}(\cdot)$. Following the scaled-form ADMM in (5)-(6), and omitting terms independent of the (x, u, z) , we define the augmented cost functional (7) with the quadratic penalty term $\tilde{\mathcal{L}}(t, x, u, y, z)$ in (8), where $y \in L^2([t_0, t_f]; \mathbb{R}^m)$ denotes the scaled dual trajectory:

$$\mathcal{J} = \Phi(t_f, x(t_f)) + \int_{t_0}^{t_f} [\tilde{\mathcal{L}}(t, x, u, y, z) + I_{\mathcal{C}}(z)] dt, \quad (7)$$

$$\tilde{\mathcal{L}}(t, x, u, y, z) = \mathcal{L}(t, x, u) + \frac{\rho}{2} \|g(t, x, u) - z + y\|^2. \quad (8)$$

The resulting ADMM iterations consist of the following

three steps. For fixed (z^k, y^k) , the primal update solves a nonlinear OCP with the augmented running cost (8), subject to the dynamics and initial condition in (1):

$$(x^{k+1}, u^{k+1}) = \arg \min_{x,u} \mathcal{J}_k(x, u) \quad (9)$$

s.t. $\dot{x} = f(t, x, u)$, $x(t_0) = x_0$,

where $\mathcal{J}_k(x, u) := \Phi(t_f, x(t_f)) + \int_{t_0}^{t_f} \tilde{\mathcal{L}}(t, x, u, y^k, z^k) dt$.

Next, the auxiliary trajectory is updated by the L^2 projection onto $\mathcal{Z}$,

$$z^{k+1} = \Pi_{\mathcal{Z}}(g^{k+1} + y^k), \quad (10)$$

where $g^{k+1}(t) := g(t, x^{k+1}(t), u^{k+1}(t))$. For a closed convex set $\mathcal{C}$, the corresponding set $\mathcal{Z}$ is closed and convex, and its L^2 projection is separable in time. In particular, $[\Pi_{\mathcal{Z}}(v)](t) = \Pi_{\mathcal{C}}(v(t))$ almost everywhere. Hence, (10) can be evaluated pointwise as $z^{k+1}(t) = \Pi_{\mathcal{C}}(g^{k+1}(t) + y^k(t))$ almost everywhere.

Finally, the scaled dual trajectory is updated via

$$y^{k+1} = y^k + g^{k+1} - z^{k+1}. \quad (11)$$

The projection and dual updates are equalities in L^2 and are therefore understood to hold almost everywhere on $[t_0, t_f]$.

3.2 Primal Update via PINN

3.2.1 PINN for TPBVP

In (9), the primal update is a continuous-time nonlinear optimal control subproblem. We solve this subproblem using an indirect optimal control formulation. Specifically, using the augmented running cost in (8), we define the augmented Hamiltonian as

$$\tilde{H}(t, x, u, \lambda; y, z) = \tilde{\mathcal{L}}(t, x, u, y, z) + \lambda^\top f(t, x, u), \quad (12)$$

where $\lambda(t)$ denotes the costate associated with the system dynamics.

Applying PMP to (12) yields the first-order necessary conditions in (13), consisting of the state equation, the costate equation, and the stationarity condition with respect to the control.

$$\begin{aligned} \dot{x} &= \frac{\partial \tilde{H}}{\partial \lambda} = f(t, x, u) \\ -\dot{\lambda} &= \frac{\partial \tilde{H}}{\partial x} = \frac{\partial \mathcal{L}}{\partial x} + \left(\frac{\partial f}{\partial x}\right)^\top \lambda + \rho \left(\frac{\partial g}{\partial x}\right)^\top (g - z + y) \\ 0 &= \frac{\partial \tilde{H}}{\partial u} = \frac{\partial \mathcal{L}}{\partial u} + \left(\frac{\partial f}{\partial u}\right)^\top \lambda + \rho \left(\frac{\partial g}{\partial u}\right)^\top (g - z + y) \end{aligned} \quad (13)$$

Together with the boundary conditions, (13) defines a TPBVP for the primal variables $(x(t), \lambda(t), u(t))$. We solve this TPBVP using a PINN by approximating (x, λ, u) using physics-informed function approximators and minimizing the residuals of (13) over collocation points on $[t_0, t_f]$.

3.2.2 Chebyshev NN with Constrained Expression

We parameterize the trajectories using a single-layer Chebyshev neural architecture with Chebyshev polynomials of the first kind as fixed hidden features:

$$\begin{aligned} T_0(x) &= 1, & T_1(x) &= x, \\ T_{n+1}(x) &= 2xT_n(x) - T_{n-1}(x). \end{aligned} \quad (14)$$

The time domain $t \in [t_0, t_f]$ is linearly mapped to the standard interval $\tau \in [-1, 1]$ via

$$\tau(t) = 2 \frac{t-t_0}{t_f-t_0} - 1, \quad \frac{d\tau}{dt} = \frac{2}{t_f-t_0}. \quad (15)$$

Using $T_i(\tau)$ as basis functions, a free function $\psi(t)$ is expanded as

$$\psi(t) = \sum_{i=0}^{L-1} \xi_i T_i(\tau(t)), \quad (16)$$

where L is the number of basis functions and ξ denotes the trainable coefficients.

We use independent free-function parameterizations $\psi_x(t)$, $\psi_\lambda(t)$, and $\psi_u(t)$ for $x(t)$, $\lambda(t)$, and $u(t)$, with coefficient vectors $\xi_x \in \mathbb{R}^{n_x L}$, $\xi_\lambda \in \mathbb{R}^{n_\lambda L}$, and $\xi_u \in \mathbb{R}^{n_u L}$. To analytically enforce boundary conditions, we adopt the constrained expression (CE) framework of the Theory of Functional Connections (TFC) [8]. For example, when the state satisfies the endpoint constraints $x(t_0) = x_0$ and $x(t_f) = x_{\text{des}}$, we construct $x(t)$ using the following constrained expression:

$$x(t) = \psi_x(t) + \frac{t_f-t}{t_f-t_0}(x_0 - \psi_x(t_0)) + \frac{t-t_0}{t_f-t_0}(x_{\text{des}} - \psi_x(t_f)).$$

This expression satisfies both boundary conditions for any choice of $\psi_x(\cdot)$. Differentiating the above with respect to time yields:

$$\dot{x}(t) = \dot{\psi}_x(t) - \frac{1}{t_f-t_0}(x_0 - \psi_x(t_0)) + \frac{1}{t_f-t_0}(x_{\text{des}} - \psi_x(t_f)).$$

The derivative of the free function is obtained using the chain rule:

$$\dot{\psi}(t) = \dot{\tau}(t) \sum_{i=0}^{L-1} \xi_i \frac{dT_i}{d\tau}(\tau(t)) = \frac{2}{t_f-t_0} \sum_{i=0}^{L-1} \xi_i T'_i(\tau(t)).$$

The same construction is applied to the costate, with the CE formed to satisfy the transversality conditions implied by the terminal cost and terminal constraints.

3.2.3 Residual Collocation and Least-Squares Update

We evaluate the TPBVP conditions in (13) at a set of collocation points $\{t_i\}_{i=1}^N \subset [t_0, t_f]$ and define the pointwise PMP residual at t_i as follows:

$$r_i(\xi_x, \xi_\lambda, \xi_u; y^k, z^k) := \begin{bmatrix} \dot{x}(t_i) - f(t_i, x, u) \\ \dot{\lambda}(t_i) + \frac{\partial \tilde{H}}{\partial x}(t_i, x, u, \lambda) \\ \frac{\partial \tilde{H}}{\partial u}(t_i, x, u, \lambda) \end{bmatrix} \quad (17)$$

Stacking these residuals yields $r(\xi_x, \xi_\lambda, \xi_u; y^k, z^k)$, and the primal update is formulated as the nonlinear least-squares problem:

$$\text{minimize}_{\xi_x, \xi_\lambda, \xi_u} \|r(\xi_x, \xi_\lambda, \xi_u; y^k, z^k)\|^2. \quad (18)$$

We solve (18) using an iterative least-squares procedure in the inner loop. Let $J(\xi) := \partial r / \partial \xi$ denote the Jacobian of the residual with respect to the concatenated coefficient vector $\xi := [\xi_x^\top, \xi_\lambda^\top, \xi_u^\top]^\top \in \mathbb{R}^p$. Since the residual is evaluated at N collocation nodes with d_r components per node, the Jacobian has the structure $J(\xi) \in \mathbb{R}^{N d_r \times p}$. At each inner iteration, we compute the Levenberg–Marquardt step

$$(J^\top J + \epsilon I) \Delta \xi = -J^\top r, \quad \xi \leftarrow \xi + \Delta \xi. \quad (19)$$

Here, $\epsilon > 0$ is a damping parameter that improves numerical stability. In our implementation, $J(\xi)$ is computed

via automatic differentiation

The PI-ADMM schematic is summarized in Fig. 1.

3.3 Consistency Analysis

We analyze fixed points of the idealized function-space formulation with exact satisfaction of the augmented first-order conditions. The implementation instead uses one inner step per outer iteration.

Proposition 1 (Fixed-Point Consistency). *Assume that $\mathcal{C}$ is nonempty, closed, and convex and that the problem data are continuously differentiable in the relevant variables. Let $(x^*, u^*, \lambda^*, z^*, y^*)$ be a fixed point of (10)–(11) satisfying (13) almost everywhere, together with the prescribed boundary and transversality conditions, where x^*, λ^* are absolutely continuous and $y^*, z^* \in L^2$. Define the path multiplier by $\mu^* := \rho y^*$. Then $(x^*, u^*, \lambda^*, \mu^*)$ satisfies: (i) $g(t, x^*(t), u^*(t)) \in \mathcal{C}$ (primal feasibility), (ii) $\mu^*(t) \in \mathcal{N}_{\mathcal{C}}(g(t, x^*(t), u^*(t)))$ (dual feasibility), and (iii) the PMP conditions in (2). If $\mathcal{C} = \mathbb{R}^m$, complementary slackness holds almost everywhere in t .*

Proof. The z -update $z^{k+1} = \arg \min_z I_{\mathcal{C}}(z) + \frac{\rho}{2} \|z - (g(t, x^{k+1}, u^{k+1}) + y^k)\|^2$ is convex in z , hence its first-order optimality condition is the subdifferential inclusion:

$$0 \in \partial I_{\mathcal{C}}(z^{k+1}) + \rho(z^{k+1} - (g(t, x^{k+1}, u^{k+1}) + y^k)). \quad (20)$$

Since $\mathcal{C}$ is closed and convex, $\partial I_{\mathcal{C}}(z) = \mathcal{N}_{\mathcal{C}}(z)$, and (20) is equivalent to:

$$\rho(g(t, x^{k+1}, u^{k+1}) - z^{k+1} + y^k) \in \mathcal{N}_{\mathcal{C}}(z^{k+1}). \quad (21)$$

At a fixed point, the y -update (11) implies $g^*(t) = z^*(t)$ almost everywhere in t , and therefore $g^*(t) \in \mathcal{C}$, proving primal feasibility. Substituting $g^* = z^*$ into (21) yields $\rho y^*(t) \in \mathcal{N}_{\mathcal{C}}(g^*(t))$, hence $\mu^*(t) \in \mathcal{N}_{\mathcal{C}}(g^*(t))$, proving dual feasibility. If $\mathcal{C} = \mathbb{R}^m$, the normal-cone condition implies the componentwise complementarity slackness almost everywhere in time.

Finally, evaluating the augmented PMP conditions (13) at the fixed point and using $g^* - z^* = 0$, the augmented constraint terms in both the costate and stationary conditions reduce to the original PMP multiplier terms $(\partial g / \partial x)^\top \mu^*$ and $(\partial g / \partial u)^\top \mu^*$, respectively. Hence, the fixed point satisfies the state equation, costate equation, and stationarity condition in (2), which coincide with the stated first-order stationarity conditions of the original constrained OCP, completing the proof. $\square$

The following corollary relates the inexact primal-step residual to the original first-order conditions.

Corollary 2 (Inexact primal-step consistency). *Assume that $\mathcal{C}$ is nonempty, closed, and convex, $\rho > 0$ is fixed, and g_x, g_u are uniformly bounded along the iterates. All residuals below are stacked over the collocation nodes. Let $r_p^{k+1} := g^{k+1} - z^{k+1}$, $r_d^{k+1} := z^{k+1} - z^k$, and $\mu^{k+1} := \rho y^{k+1}$. If $\|r(\xi^{k+1}; y^k, z^k)\|_2 \leq \varepsilon_k$, then there exists a constant $c_g > 0$ such that*

$$\|e_{\text{PMP}}^{k+1}\|_2 \leq \varepsilon_k + c_g \|r_d^{k+1}\|_2, \quad (22)$$

where e_{PMP}^{k+1} is the residual of (2) evaluated with μ^{k+1} . The projection and dual updates further yield $\mu_i^{k+1} \in \mathcal{N}_{\mathcal{C}}(z_i^{k+1})$ and $\text{dist}(g_i^{k+1}, \mathcal{C}) \leq \|r_{p,i}^{k+1}\|_2$ at each node.

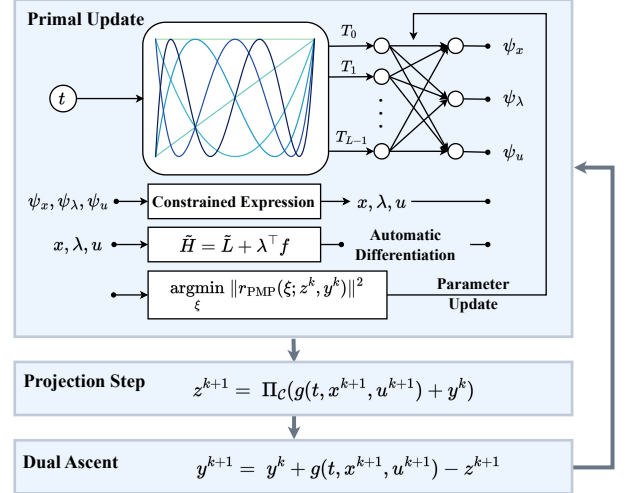


Figure 1: Schematic of the proposed PI-ADMM framework for path-constrained continuous-time optimal control.

Proof. The dual update gives $g^{k+1} - z^k + y^k = y^{k+1} + r_d^{k+1}$. Thus, the augmented and original PMP residuals differ by $\rho g_x^\top r_d^{k+1}$ and $\rho g_u^\top r_d^{k+1}$, yielding (22). The remaining claims follow from projection optimality. $\square$

To bound the residual between collocation nodes, define $e_{\text{PMP}}^{k+1}(t)$ using the parameterized trajectories and piecewise-linear interpolation of nodal multipliers on a grid containing both endpoints. If it is Lipschitz with constant $L_{e,k}$, then

$$\|e_{\text{PMP}}^{k+1}\|_{L_\infty} \leq \varepsilon_k + c_g \|r_d^{k+1}\|_2 + L_{e,k} h_N, \quad (23)$$

where $h_N := \sup_{t \in [t_0, t_f]} \min_i |t - t_i|$.

Remark 3 (Scope of the Convex-Set Assumption). *The identity $\partial I_{\mathcal{C}} = \mathcal{N}_{\mathcal{C}}$ used above relies on the closed convex structure of $\mathcal{C}$. In practice, the proposed framework can also be applied to nonconvex feasible sets using projections that may be set-valued or expansive. The fixed-point consistency result above does not directly extend to this setting, although nonconvex projection-based methods have shown stable practical performance [17–20].*

4 Numerical Experiments

4.1 Problem Definition

We evaluate the proposed PI-ADMM framework on a 3-DoF PDG problem [5, 17]:

$$\begin{aligned} \text{minimize}_{u(\cdot), x(\cdot)} \quad & -m(t_f) + w_{\text{slew}} \int_{t_0}^{t_f} \|u(t)\|^2 dt \end{aligned} \quad (24a)$$

$$\text{subject to} \quad \dot{r} = v, \quad \dot{v} = T/m + g_{\text{grav}}, \quad (24b)$$

$$\dot{m} = -\alpha \|T\|, \quad \dot{T} = u, \quad (24c)$$

$$\|v\| \leq v_{\text{max}}, \quad \|u\| \leq \dot{T}_{\text{max}}, \quad (24d)$$

$$m_{\text{dry}} \leq m \leq m_0, \quad (24e)$$

$$T_u \geq \cos(\gamma_p) \|T\|, \quad (24f)$$

$$r_u \geq \tan(\gamma_{\text{gs}}) \|r_{e:n}\|, \quad (24g)$$

$$T_{\text{min}} \leq \|T\| \leq T_{\text{max}}, \quad (24h)$$

$$r(t_0) = r_0, \quad v(t_0) = v_0, \quad m(t_0) = m_0, \quad (24i)$$

$$r(t_f) = r_{\text{des}}, \quad v(t_f) = v_{\text{des}}. \quad (24j)$$

Table 1: Nominal problem data for the 3-DoF PDG benchmark used in the numerical study.

Parameter	Value	Parameter	Value
t_f	35 s	N	50
g_{grav}	9.81 m/s ²	I_{sp}	270 s
T_{min}	40 kN	T_{max}	60 kN
$\dot{T}_{\text{max}}$	0.05 T_{max}/s	v_{max}	35 m/s
γ_p	10°	γ_{gs}	30°
m_0	5000 kg	m_{dry}	3000 kg
r_0	[800, 250, 100] ^T m	r_{des}	[10, 0, 0] ^T m
v_0	[-8, 10, 0] ^T m/s	v_{des}	[0, 0, 0] ^T m/s

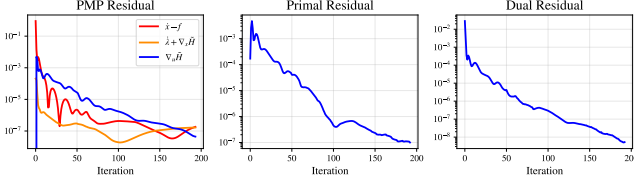


Figure 2: Outer-iteration residual histories of the inexact PI-ADMM solver on the nominal 3-DoF PDG benchmark. The plotted quantities are the residual of the augmented PMP least-squares problem (18), the ADMM primal residual $\|g(t, x^k, u^k) - z^k\|$, and the ADMM dual residual $\|z^k - z^{k-1}\|$.

To explicitly model thrust slew rate limits, we augment the translational dynamics with the thrust vector as a state and treat the slew as the control input. The state and control are defined as $x = [r^\top, v^\top, m, T^\top]^\top \in \mathbb{R}^{10}$ and $u = \dot{T} \in \mathbb{R}^3$, where $r \in \mathbb{R}^3$ is position, $v \in \mathbb{R}^3$ is velocity, $m \in \mathbb{R}$ is mass, and $T \in \mathbb{R}^3$ is the thrust vector. The terminal cost maximizes the terminal mass, while the running cost penalizes the thrust slew, with w_{slew} denoting the associated weighting parameter. The dynamics (24b)–(24c) describe translational motion under thrust and gravity, propellant depletion, and the slew dynamics.

The path constraints enforce velocity, slew-rate, and dry-mass limits (24d)–(24e), together with thrust pointing and glide-slope constraints (24f)–(24g). The thrust-magnitude constraint (24h) produces a feasible nonconvex annulus set, which can induce an expansive projection (Remark 3). The boundary conditions are given in (24i)–(24j). Since the initial and terminal thrust are free and Φ depends on $m(t_f)$, the corresponding transversality conditions are:

$$\lambda_T(t_0) = 0, \quad \lambda_T(t_f) = 0, \quad \lambda_m(t_f) = -1. \quad (25)$$

These transversality conditions are incorporated when constructing the constrained expressions for the costates in the primal TPBVP solver.

4.2 Simulation Results

The simulation setup is summarized in Table 1. Here, K denotes the number of time nodes, and α in the mass dynamics (24c) is the propellant mass flow coefficient, given by $\alpha = 1/(I_{\text{sp}}g_{\text{grav}})$. The proposed PI-ADMM solver is compared with two baselines: a nonlinear programming (NLP) approach implemented in CasADi and solved by IPOPT, and a lossless convexification (LCvx) formulation implemented in CVXPY and solved by MOSEK [2, 5].

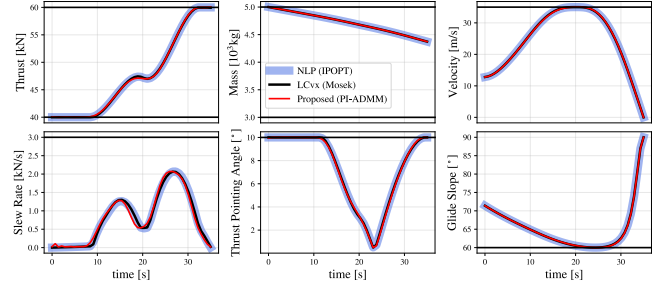


Figure 3: Constraint-relevant state and control histories for the nominal 3-DoF PDG benchmark obtained by IPOPT, LCvx, and PI-ADMM.

Table 2: Quantitative performance comparison on the benchmark.

Method	Propagation Error	Fuel Consumption	Constraint Violation
NLP (IPOPT)	0.109	626.52	1.18×10^{-6}
LCvx (MOSEK)	0.109	626.53	1.72×10^{-6}
Proposed (PI-ADMM)	0.068	626.46	2.30×10^{-6}

*Propagation error denotes the state mismatch obtained by roll-out of the returned solution, fuel consumption is computed as $m_0 - m(t_f)$, and constraint violation denotes the maximum path-constraint violation over the horizon.

Each outer iteration uses one Levenberg–Marquardt step. Successful termination requires the augmented PMP, primal, and dual residual norms to satisfy their respective tolerances. Fig. 2 shows that all three residual measures decrease by several orders of magnitude, indicating improved primal-dual consistency over the iterations.

Fig. 3 compares the state and control profiles obtained by the proposed method and the baseline solvers. All three methods produce closely matching trajectories, and the proposed PI-ADMM solution exhibits small constraint violations at the evaluated time points. The close agreement in the trajectory profiles indicates that the projection-based constraint handling remains effective even in the presence of the nonconvex thrust-magnitude constraint. This qualitative agreement is also reflected in the quantitative metrics reported in Table 2. The three methods yield nearly identical fuel consumption and similarly small constraint violations, all on the order of 10^{-6} . The propagation errors are also comparable, with the proposed PI-ADMM method achieving the smallest value among the tested solvers. These results indicate that the proposed method attains solution quality comparable to both baseline methods.

Fig. 4 compares wall-clock solve times over 50 runs per setting, with the number of temporal nodes ranging from 40 to 400. PI-ADMM achieves the lowest mean solve time across most of the tested node counts, and its measured solve time varies only weakly with node count over this range.

This observed behavior is consistent with the coefficient-space structure of the primal solver. For a fixed basis size, increasing the number of collocation nodes enlarges the residual vector and Jacobian but leaves the dimension of the LM linear system unchanged. Residual and Jacobian evaluation, as well as the formation of the normal equa-

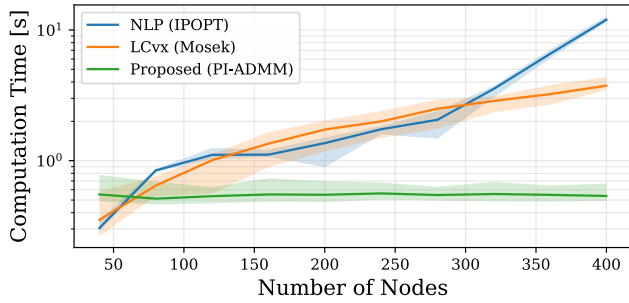


Figure 4: Wall-clock solve-time scaling with temporal discretization size for the nominal 3-DoF PDG benchmark. For each node count, 50 repeated runs are performed for IPOPT, LCvx, and PI-ADMM; solid lines indicate the empirical mean and shaded bands indicate the range.

tions, still incur node-dependent costs; nevertheless, the measured results demonstrate favorable computational scaling on this benchmark.

5 Conclusion

This paper presented PI-ADMM, a physics-informed ADMM framework for constrained continuous-time optimal control, and applied it to a 3-DoF PDG problem. By combining an augmented PMP-based primal update with projection-based ADMM constraint handling, the method preserves PMP structure while effectively enforcing path feasibility. A fixed-point analysis establishes consistency with the first-order stationarity conditions of the original problem under convexity and regularity assumptions. Numerical results showed solution quality comparable to IPOPT and LCvx in fuel consumption, constraint satisfaction, and propagation error, while achieving the lowest mean solve time across most tested node counts. Overall, these results demonstrate that PI-ADMM provides a practical alternative to full direct transcription for handling complex path constraints in indirect optimal control.

A key strength of the framework is its coefficient-space primal update, whose LM linear system remains fixed in dimension under node refinement for a fixed basis size. This structure suggests potential for onboard guidance applications such as planetary landing and reusable launch vehicle recovery. Its limitations include reliance on empirical parameter tuning and restriction of the fixed-point consistency result to the closed convex-set setting, while a general guarantee for nonconvex projection-based updates remains unavailable. Future work includes extension to 6-DoF dynamics, free-final-time problems, and theoretical analysis of nonconvex projection-based updates.

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